

EE3621: Digital Signal Processing

Lecture 30: DSP Capstone — System Design, Implementation & Future Trends (III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** System Design Methodology: Requirements, algorithmic choices, fixed vs floating point.
 - **05:00 – 12:00 (7 mins):** Fixed-Point DSP Implementation: Q-format notation, overflow, scaling, and rounding strategies.
 - **12:00 – 18:00 (6 mins):** DSP Processor Architecture: Harvard architecture, MAC, barrel shifter, circular buffers, and SIMD.
 - **18:00 – 22:00 (4 mins):** Hardware Trade-offs (FPGA vs DSP vs GPU) & Real-Time System Design.
 - **22:00 – 30:00 (8 mins):** Course Review: Interconnections of key topics.
 - **30:00 – 35:00 (5 mins):** Emerging Trends (TinyML, QSP) & Career Paths in DSP.
 - **35:00 – 40:00 (5 mins):** Comprehensive Final Exam Problems discussion.
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2 System Design Methodology

Designing a DSP system requires traversing from abstract mathematics to physical hardware.

2.1 Requirements Analysis

Before writing any code, engineers must define: 1. **Sampling Rate (f_s):** Dictates the computational budget. 2. **Dynamic Range & SNR:** Determines the word length. 3. **Latency:** Critical for real-time control loops. 4. **Power Consumption:** Dictates hardware selection.

2.2 Algorithm Selection

FIR vs. IIR: FIR provides linear phase and unconditional stability but requires more MAC operations. IIR requires fewer coefficients but is prone to limit cycles.

2.3 Floating-Point vs. Fixed-Point

Floating-point is easier to program with vast dynamic range but takes more power/area. Fixed-point is harder to program (requires manual scaling) but offers low power and high speed.

3 Fixed-Point DSP Implementation

3.1 Two's Complement Arithmetic

Most DSPs use two's complement. An N -bit number represents values from -2^{N-1} to $2^{N-1} - 1$.

3.2 Q-Format Notation

To represent fractional numbers in integer hardware, we use $Q(m.n)$ notation:

- m : Number of integer bits (including sign bit).
- n : Number of fractional bits.
- Total bits: $N = m + n$.
- Range: $[-2^{m-1}, 2^{m-1} - 2^{-n}]$

3.3 Overflow Handling

When additions exceed the maximum value, two's complement wraps around. **Saturation Arithmetic** clamps the result to the maximum/minimum representable value instead of wrapping.

3.4 Scaling and Rounding

When multiplying two $Q(m.n)$ numbers, the result is in $Q(2m.2n)$. To store this back, we must shift right by n bits and round. Truncation causes a negative DC bias. Rounding (adding 2^{n-1} before truncating) provides zero-mean error.

3.5 Block Floating-Point

A block of data shares a single exponent. The block is scanned for the maximum magnitude, and all values are scaled to maximize dynamic range.

4 DSP Processor Architecture

4.1 Harvard Architecture

Separate memory spaces and buses for instructions and data. Allows the processor to fetch an instruction and read/write data simultaneously.

4.2 Single-Cycle MAC

The Multiply-Accumulate operation executes in a single cycle using a dedicated hardware multiplier and a wide accumulator.

4.3 Barrel Shifter

Can shift a data word left or right by any number of bits in a single clock cycle. Crucial for scaling.

4.4 Circular Buffers

FIR filters use a delay line. A circular buffer updates pointers modulo the buffer size, implementing delays with zero data-movement overhead.

4.5 SIMD Extensions

Single Instruction, Multiple Data allows one instruction to operate on multiple data points simultaneously.

5 FPGA vs DSP Processor vs GPU

Feature	FPGA	DSP Processor	GPU
Architecture	Reconfigurable logic	Programmable CPU	Massive SIMD
Parallelism	True hardware	Instruction level	Thread-level
Latency	Extremely low	Low	High
Power Eff.	Very high	High	Low

Table 1: Hardware Trade-offs

6 Real-Time System Design

6.1 Interrupt vs DMA

DMA (Direct Memory Access): Hardware writes ADC samples directly to memory. CPU is only interrupted when a full block is ready.

6.2 Double-Buffering (Ping-Pong)

While the CPU processes Buffer A, the DMA fills Buffer B. Hides I/O latency.

6.3 Worst-Case Execution Time (WCET)

The WCET must be strictly less than the block processing deadline.

7 Course Review — Key Connections

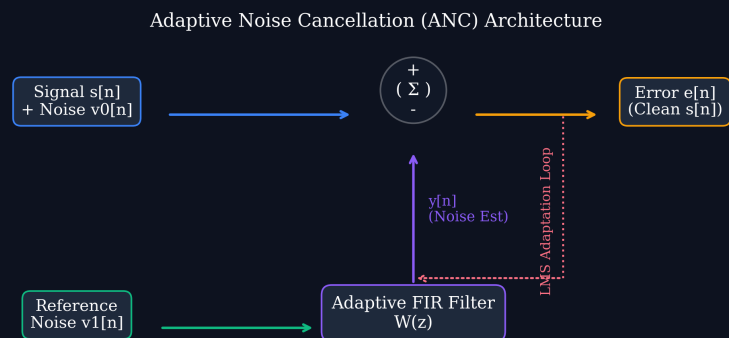


Figure 1: Adaptive Noise Cancellation (ANC) architecture.

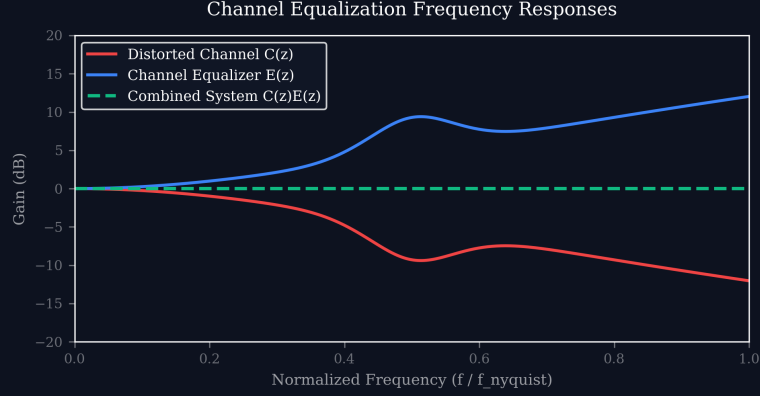


Figure 2: Channel Equalizer frequency responses.

8 Emerging Trends & Career Paths

Trends: Deep Learning on Edge (TinyML), Neuromorphic Computing, Quantum Signal Processing, Software-Defined Radio.

Careers: RF/Communications, Embedded Audio/Video, Power Electronics, Biomedical, Autonomous Vehicles.

9 Comprehensive Final Exam Problems

Problem 1: Filter Design & Quantization

Design a 1st-order IIR lowpass filter with a pole at $z = 0.8$. Implement it in $Q(2.14)$.

Solution: Difference equation is $y[n] = x[n] + 0.8y[n-1]$. $0.8 \times 2^{14} \approx 13107$. Truncation error variance $\sigma_e^2 = \frac{(2^{-14})^2}{12} \approx 3.1 \times 10^{-10}$. Output noise variance $\sigma_y^2 = \sigma_e^2 \frac{1}{1-0.8^2} = 8.6 \times 10^{-10}$.

Problem 2: Adaptive Filtering

An LMS filter with $M = 2$ taps. Input $x[n]$ is white noise with variance 1. Find max step size μ .

Solution: $\mathbf{R} = \mathbf{I}$. $\text{Tr}(\mathbf{R}) = 2$. Stability condition $0 < \mu < \frac{1}{\text{Tr}(\mathbf{R})} \implies 0 < \mu < 0.5$.

Problem 3: Multirate System Design

Upsample 8 kHz to 48 kHz.

Solution: Interpolation factor $L = 6$. Insert 5 zeros. Design lowpass $H(z)$ with cutoff $\pi/6$. Decompose into 6 polyphase components to compute efficiently at 8 kHz.

10 Key Formulas Summary

Concept	Formula
Q-format Range	$[-2^{m-1}, 2^{m-1} - 2^{-n}]$
LMS Update	$\mathbf{w}[n+1] = \mathbf{w}[n] + 2\mu e[n]\mathbf{x}[n]$
LMS Stability	$0 < \mu < \frac{1}{\text{Tr}(\mathbf{R})}$
Zero-Forcing	$E(z) = \frac{1}{C(z)}$
Polyphase Filter	$H(z) = \sum_{k=0}^{L-1} z^{-k} E_k(z^L)$

Table 2: Key Formulas

11 Checkpoint Questions

1. **Q1:** Why is a barrel shifter essential in a fixed-point DSP processor?

Answer: Multiplying $Q(m.n)$ numbers gives a $Q(2m.2n)$ product. To store it back, we shift right by n bits. A barrel shifter does this in one cycle.

2. **Q2:** Compare truncation vs rounding.

Answer: Truncation discards lower bits, introducing a negative DC bias. Rounding adds 2^{n-1} before truncating, providing a zero-mean error.

3. **Q3:** How does double-buffering prevent data loss?

Answer: DMA fills Buffer A while CPU processes Buffer B. They swap roles when Buffer A is full, decoupling continuous I/O from bursty CPU processing.